Behavioural Economics

ECO1028 – Politics Without Romance

Beatriz Gietner
beatriz.gietner@dcu.ie

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Outline

The Rational Agent: A Quick Recap

What is Behavioural Economics?

Non-Standard Preferences

Non-Standard Beliefs

Nudge Theory and Libertarian Paternalism

Why Does This Course Cover Behavioural Economics? i

We are near the end of the course. Look at what we have built.

- **Downs:** voters are rationally ignorant; not worth learning about politics
- **Median voter:** parties rationally converge on the pivotal voter's preferences
- **Political business cycles:** politicians rationally time fiscal policy around elections
- **Niskanen:** bureaucrats rationally maximise their agency's budget
- **Tiebout:** citizens rationally sort across jurisdictions by revealed preference
- **Olson, Tullock, Stigler:** interest groups rationally lobby; industries rationally capture regulators

Why Does This Course Cover Behavioural Economics? ii

Every single model runs on the same assumption: actors know what they want, process information correctly, and respond to incentives predictably.

Today we ask: what happens to all of this if that assumption is wrong - not randomly, but *systematically* and *predictably*?

Three Connections Worth Keeping in Mind

As we go through the lecture, watch for these threads:

1. **Loss aversion may amplify Olson.**

   Concentrated losers from a reform do not just have stronger *incentives* to lobby. Behavioural evidence suggests that losses are often felt more sharply than equivalent gains. That psychological asymmetry can compound the organisational one.

2. **Present bias explains why good long-run policy is politically impossible.**

   Voters are present-biased, and so are politicians facing election cycles. Reforms that cost now and pay off in five years are doubly cursed: the costs are felt immediately, and both the electorate and the government discount the future too steeply. This shifts the interpretation of the political business cycle from pure opportunism to structural forces.

3. **The hardest question: if citizens are biased, so are voters - so who corrects it?**

   This is the capstone problem of the course. Public choice says governments fail because politicians are self-interested. Behavioural economics says we may need intervention because citizens are biased. But biased citizens elect politicians who *exploit* those biases. We return to this at the end - and there is no clean answer.

The Tension at the Heart of This Lecture i

Public Choice argument

Governments fail.
Politicians respond to incentives, not to the public good.
Regulation gets captured.
Intervention makes things worse.

Conclusion: leave people alone to make their own choices.

Behavioural argument

Markets fail too.
People are systematically biased.
Firms exploit those biases profitably.
Some intervention can improve welfare without coercion.

Conclusion: design the choice environment more carefully.

Glaeser's (2006) objection: governments are staffed by the same biased, present-biased, overconfident humans as everyone else - plus they face additional political distortions. Why would we expect them to correct citizen biases rather than exploit them?

The Tension at the Heart of This Lecture ii

This is the unresolved tension we will carry through today's lecture.

There is no clean answer. The goal is to be able to articulate both sides precisely.

The Rational Agent: A Quick Recap

The Character at the Centre of Economics i

Every model we have studied this semester contains the same character.

He is called homo economicus - economic man.

He has some very convenient properties:

- He knows what he wants, and his preferences don't change depending on his mood
- He processes all available information correctly
- He makes plans and follows through on them - always
- He responds to incentives in predictable ways
- He is entirely self-interested

The Character at the Centre of Economics ii

He is enormously useful. The Olson model works. The Tullock model works. The Stigler model works. All of them run on this character.

Today's question: what happens when we replace him with an actual human being?

Where the Model Works - and Where It Doesn't

The rational agent model works best when:

- People make the same type of decision repeatedly - they learn from mistakes
- The stakes are high enough to pay attention
- The choice is simple and the information is clear
  - ▷ Buying coffee, haggling at a market, bidding on a stock

It struggles when:

- The decision is rare and high-stakes - choosing a pension, buying a house
- The cost comes now but the benefit comes much later - or vice versa
- The relevant information is hidden or hard to process
  - ▷ Exactly the decisions that matter most in people's lives

Golden rule: remember your assumptions.

Evaluate the model by whether its simplifications affect the problem at hand, rather than by whether it is literally "true".

What is Behavioural Economics?

Before We Begin: Three Quick Questions

Raise your hand if...

1. ...you have ever paid for a gym membership and not gone.
2. ...you have ever told yourself "I'll start saving next month" - and then not done it.
3. ...you have ever felt *more* upset about losing €20 than happy about finding €20.

If any of these apply to you, you are not irrational.

You are **predictably human** - and that is exactly what behavioural economics studies.

What is Behavioural Economics?

Standard economics asks: given that people are rational, what will they do?

Behavioural economics asks: given how people *actually* think and decide, what will they do - and what should policy do about it?

The field draws on psychology to modify the assumptions of the rational agent model while keeping the framework intact.

The key word is predictable. In behavioural economics we identify systematic patterns of deviation - the same mistakes, made by the same types of people, in the same types of situations.

If the mistakes were random, they would cancel out. Because they are systematic, they matter for markets, institutions, and policy.

Key Definitions

- **Behavioural economics:** the study of how psychological factors shape economic decisions - and what to do about it in policy design.
- **Bounded rationality:** the idea that people use simplified rules (heuristics) because thinking carefully is costly and time-consuming.
- **Heuristic:** a mental shortcut. Usually good enough; sometimes systematically wrong.
- **Cognitive bias:** a predictable pattern of error in judgement or decision-making.
- **Present bias:** the tendency to over-weight costs and benefits that happen *now* relative to ones in the future.
- **Loss aversion:** the tendency to feel losses more strongly than equivalent gains.
- **Status quo bias:** the tendency to stick with whatever is already in place.
- **Nudge:** a change to the way choices are presented that predictably shifts behaviour, without removing options and without *significantly* changing economic incentives.
- **Choice architecture:** the design of the environment in which decisions are made.

Three Ways People Deviate from the Standard Model

We can organise the findings into three buckets:

1. **Non-standard preferences**

   People want things the rational agent model says they shouldn't want. They struggle to follow through on their own plans. They care about fairness, not just their own payoff. They feel losses more than gains.

2. **Non-standard beliefs**

   People hold *systematically wrong* views about the world. They are overconfident about their own abilities. They see patterns in randomness. They assume tomorrow will feel just like today.

3. **Non-standard decision-making**

   Even with the right values and beliefs, people make predictable errors. They ignore information that isn't right in front of them. They are swayed by irrelevant factors - the order of options, what others are doing, whether they're in a good mood.

What We Will Learn

By the end of this lecture, you should be able to explain:

- Why people under-save, over-eat, and overpay - and why blaming individual weakness misses the point
- Why losing €50 feels worse than finding €50 feels good - and what this implies for wages, contracts, and housing policy
- Why people are systematically overconfident and how this shapes everything from mergers to financial crises
- Why the framing and presentation of a choice can matter as much as the choice itself
- What nudges are, why they work, and what the limits of "libertarian paternalism" are

Core readings:

- DellaVigna (2007), "Psychology and Economics: Evidence from the Field"
- Thaler & Sunstein (2003), "Libertarian Paternalism"

Non-Standard Preferences

The Problem of Self-Control

Imagine it's Sunday evening.

You plan to go for a run on Monday morning, eat well, and start that assignment you've been putting off.

Monday arrives. You sleep in. You eat cereal at noon. The assignment waits.

Irrationality? No, it is time inconsistency.

Your Sunday-evening self and your Monday-morning self have *genuinely different preferences*. The future always looks more disciplined than the present actually is.

The rational agent model assumes these two selves are identical. Behavioural economics says this is one of the most consequential mistakes the model makes.

Present Bias: The Core Idea

We all discount the future - that's normal and rational. A reward next year is worth less to you than the same reward today.

Present bias is different. It is an *extra* over-weighting of the present - a special impatience that kicks in only when costs or benefits are *right now*.

A simple illustration:

- Most people prefer €100 today over €110 in one week
- But most people also prefer €110 in 5 weeks and 1 day over €100 in 5 weeks
- The gap is identical - one week - but the first feels much harder to wait

The present is special in a way that our plans don't account for.

For those interested in the formal model: see the Appendix for the (β, δ) framework.

Two Types of Goods

Present bias has different consequences depending on what kind of good you're dealing with.

Investment goods

Cost now, benefit later

- Going to the gym
- Saving for retirement
- Studying for exams
- Eating vegetables

Present bias makes us *do less* of these than we intend to.

Leisure goods

Pleasure now, cost later

- Junk food
- Credit card spending
- Doomscrolling at midnight
- Skipping lectures

Present bias makes us *do more* of these than we plan to.

The gap between plans and actions is systematic and exploitable.

The Gym Story

A gym near a large US university offered two contracts:

- **Monthly membership:** flat €80 fee, unlimited visits
- **Pay per visit:** €10 each time

If you're rational, you only sign up for the monthly membership if you expect to go **at least 8 times**.

What did researchers find?

- Members on monthly contracts attended on average 4.8 times per month
- Their effective cost per visit: **€17** - almost double the pay-per-visit price
- Almost none of them cancelled

Why don't they switch or cancel? On the day they signed up, future gym visits felt certain. Each Monday, the run can always start "next Monday." Cancelling requires a phone call - which also gets put off.

The Deadline Story i

An experiment with MBA students at MIT:

Students had three assignments to submit over a semester.

- **Group A:** could set their own deadlines (binding - lateness cost marks)
- **Group B:** all deadlines at the very end of semester
- **Group C:** evenly-spaced deadlines set by the instructor

Rational prediction: Group A should set all deadlines at the end - why constrain yourself?

What happened:

- 68% of Group A chose early deadlines - they *wanted* to be constrained

The Deadline Story ii

- Group A did better than Group B (self-set deadlines help)
- Group C did best of all (external structure beats self-set structure)

People use rules, deadlines, and commitments as tools against their future selves. This only makes sense if you know your future self will behave differently to your current self.

The Pension Default Story

A large US company changed one thing about its pension scheme:

Before:
Employees were not enrolled.
To join: fill in a form.
Participation rate: 49%

After:
Employees were automatically enrolled.
To leave: fill in a form.
Participation rate: 86%

The form required was *identical*. The financial incentives were *unchanged*. The only difference was which option required action.

The rational agent model predicts this should not matter. But it does matter - a lot! - because inaction is the path of least resistance when you're present-biased and can always start "next month."

Ireland's 2023 Automatic Enrolment Act is built on exactly this finding.

Discussion: Who Is Responsible?

You now know that:

- People systematically under-save for retirement
- They over-commit to gym memberships they won't use
- They procrastinate on tasks they genuinely want to complete

Three positions - which do you find most convincing?

1. **Personal responsibility:** these are individual choices and the state has no business intervening
2. **Soft intervention:** change the default, but let people opt out freely - libertarian paternalism
3. **Hard intervention:** mandatory pension contributions, sugar taxes - self-control problems create a case for coercion

Think about: does your answer change if the person experiencing the self-control problem is poor? Or elderly? Or a teenager?

Loss Aversion: Losses Loom Larger Than Gains i

Try this thought experiment:

I flip a fair coin.

- Tails: you lose €50
- Heads: you win €X

What does X have to be for you to take the bet?

Most people say €100–150. Some say more. Almost no one says €50 - even though that would make it a fair bet with positive expected value.

This is loss aversion. The pain of losing €50 is often felt more strongly than the pleasure of gaining €50. In many experimental settings, people behave as if losses carry roughly **twice** the weight of equivalent gains, though estimates vary across designs and settings.

Loss Aversion: Losses Loom Larger Than Gains ii

The core point is qualitative rather than mechanical: losses and gains are not treated symmetrically.

Reference Points: It Depends Where You Started

Loss aversion implies that how you frame an outcome matters as much as the outcome itself.

Same economic reality, different framing:

- "You will receive a €1,000 bonus" vs. "Your bonus has been cut from €2,000 to €1,000"
- "This product contains 10% fat" vs. "This product is 90% fat-free"
- "A 5% pay rise when inflation is 7%" vs. "A 2% pay cut"

The reference point is whatever you treat as the baseline for comparison. Gains and losses are evaluated relative to that point rather than to absolute wealth.

The standard model says only final outcomes matter. Loss aversion says the path to those outcomes matters enormously.

Loss Aversion in the Real World: The Mug Experiment i

A classic experiment:

Researchers randomly give half a group of students a coffee mug. They then ask:

- Sellers (those with the mug): what is the minimum price you would accept to sell it?
- Buyers (those without): what is the maximum price you would pay to buy it?

Standard prediction: sellers and buyers should value the mug roughly the same - it's the same mug.

What happened:

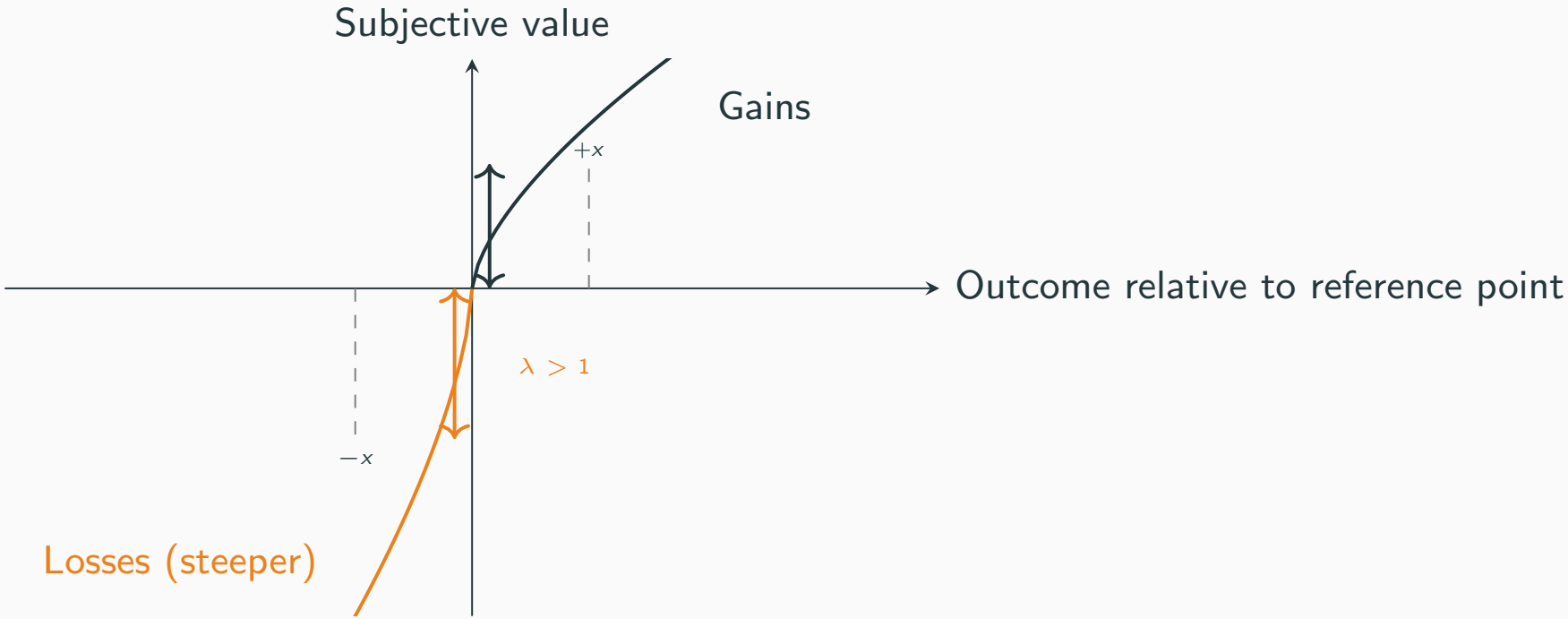
- Sellers: median asking price **$5.75**
- Buyers: median willingness to pay **$2.25**

Loss Aversion in the Real World: The Mug Experiment ii

- Ratio: roughly 2.5 to 1

The act of *owning* something changes how much you value it. Giving it up feels like a loss. Loss aversion makes you demand more to part with it than you would ever have paid to get it.

Loss Aversion: The Value Function



Illustrative prospect-theory value function. Many applied papers use loss-aversion parameters around 2, but the exact value is not universal.

Field Evidence: The Endowment Effect

Kahneman, Knetsch & Thaler (1990):

- Subjects randomly assigned a coffee mug
- **WTA** (sellers): median \$5.75
- **WTP** (buyers): median \$2.25
- Standard theory: WTA $\approx$ WTP
- Observed ratio: WTA $\approx 2.5\times$ WTP

List (2003) - Field evidence with sports memorabilia:

- Inexperienced traders: strong endowment effect (switched only 6.8% of the time)
- Experienced traders: much weaker endowment effect (switched 46.7%)

Careful interpretation: market experience appears to attenuate the *observed* endowment effect. Whether that reflects weaker loss aversion, different reference points, learning, or selection is less clear.

Field Evidence: Target Earnings in Labour Supply i

Camerer et al. (1997) - New York City Taxi Drivers:

- On high-wage days (rainy days, big events), the standard model predicts more hours
- One interpretation of the data is that some drivers behave as if they have a **daily earnings target**
- On high-demand days, some drivers appear to stop earlier once that target is reached
- This produces estimates that are hard to reconcile with the standard intertemporal labour-supply model

Fehr & Goette (2007) - Bike Messengers:

- Randomly assigned 25% wage increase for one month
- Messengers signed up for more shifts (consistent with both models)

Field Evidence: Target Earnings in Labour Supply ii

- Within shifts, effort responses are consistent with reference-dependent preferences

Takeaway: there is suggestive evidence for target earnings and reference dependence in labour supply, but the magnitude and generality of the effect remain debated.

Loss Aversion in Wages: Implications for Labour Markets i

Kahneman, Knetsch & Thaler (1986) - Survey evidence:

- 62% of respondents find a **7% nominal wage cut** unfair in zero inflation
- But only 22% find a **5% nominal rise** unfair when inflation is 12%
- In both cases, real wages fall by $\approx 7\%$ - but framing as a loss matters

Card & Hyslop (1997) - Wages data:

- Distribution of year-on-year wage changes has a spike at zero and a deficit just below
- Employers often keep wages nominally flat rather than cutting them directly

Loss Aversion in Wages: Implications for Labour Markets ii

- This pattern is consistent with workers resisting nominal wage cuts more strongly than equivalent real wage cuts delivered through inflation

Implication: downward nominal wage rigidity may partly reflect loss aversion alongside institutional factors such as bargaining, contracts, and norms. That can matter for unemployment and recession dynamics.

Social Preferences

The standard model assumes pure self-interest. A large body of evidence shows that people also care about fairness, reciprocity, and the welfare of others.

Dictator Game (Forsythe et al., 1994):

- One player allocates \$10 between themselves and a stranger
- Standard prediction: keep all \$10
- Observed: 60% transfer a positive amount; average transfer $\approx$ \$2

Gift Exchange in the Workplace (Gneezy & List, 2006):

- Workers unexpectedly told their wage was raised from \$12 to \$20/hour
- Initial productivity jumps: **+20% in hour 1**
- But the effect largely fades after 2 hours
- Key finding: the gift exchange has an emotional component that dissipates

Non-Standard Beliefs

Overconfidence

Two forms of overconfidence with strong field evidence:

1. **Overconfidence about own ability**
   - Svenson (1981): 93% of subjects rate themselves as above-median drivers
   - Camerer & Lovallo (1999): excess entry in business competitions where winners are determined by skill
2. **Overconfidence about precision of information**
   - Alpert & Raiffa (1982): 98% confidence intervals contain the true answer only 60% of the time
   - Odean (1999): individual investors trade too much; stocks sold outperform stocks bought by $\approx 3\%$ over the next year
   - Malmendier & Tate: overconfident CEOs are 55% more likely to pursue acquisitions; they destroy shareholder value

The Law of Small Numbers and Gambler's Fallacy

Tversky & Kahneman's representativeness heuristic:

People expect small samples to look like large-sample population distributions.

Two related errors:

- **Gambler's fallacy:** after a sequence of Heads, people expect Tails more. In a lottery with recent winning numbers, bettors avoid those numbers - reducing their expected payout in a pari-mutuel system (Terrell, 1994: winners pay 33% more if the number won in the last week)
- **Overinference / Hot hand fallacy:** after observing a sequence of good outcomes, people over-update that the underlying process is "trending" - 401(k) investors allocate much more to employer stock after recent good performance, even though past returns predict nothing (Benartzi, 2001)

Projection Bias

People over-project current tastes onto future preferences.

Loewenstein, O'Donoghue & Rabin (2003) model this as:

$$\hat{u}(c, s) = (1 - \alpha)\, u(c, s) + \alpha\, u(c, s_0)$$

where s_0 is the current state and $\alpha \in [0, 1]$ measures the degree of projection.

Classic example (Read & van Leeuwen, 1998):

- Workers asked to choose a snack for delivery next week
- Asked when **hungry**: 78% chose unhealthy snack
- Asked when **satiated**: 42% chose unhealthy snack
- Current hunger state is (incorrectly) projected onto future preferences

Field evidence (Conlin, O'Donoghue & Vogelsang): Cold-weather clothing ordered on colder days is returned more often - buyers over-predict how much they will value it. Structural estimate: $\hat{\alpha} \approx 0.5$.

Limited Attention i

People don't process all available information equally.

What is visible and salient gets attention; what is buried in small print, separated onto a different line, or announced on a quiet Friday does not. The same total cost can feel very different depending on how it is presented.

Key field studies:

- **eBay shipping costs** (Hossain & Morgan, 2006): shifting $3.99 from the bid price to the shipping cost - same total price - raises seller revenue by $1.79. Buyers anchor on the headline number and under-react to the rest.
- **Non-transparent taxes** (Chetty et al., 2007): when a 7.4% sales tax is printed on the price tag rather than added at the till, demand falls by 8.8%. Out of sight genuinely means out of mind.

Limited Attention ii

- **Friday earnings announcements**: stocks announced on Fridays show delayed price adjustment. Firms learn to time bad news for low-attention days.

Policy implication: making information harder to ignore is itself a policy lever. Mandatory calorie labelling, plain tobacco packaging, and price comparison websites all work by bringing hidden costs into view - no ban required.

Menu Effects and Choice Overload i

People do not always process every option carefully.

- When the choice set is large, people may rely on simple cues such as order, salience, defaults, or what stands out visually
- More options can improve welfare in principle, but in practice they can also make decisions harder
- Small design features can therefore shift choices even when the available options stay the same

Examples from the literature:

- **Ballot order effects:** candidates listed earlier can receive a modest advantage, though the size of the effect varies across settings
- **Supermarkets and restaurants:** products placed at eye level or made more salient tend to be chosen more often

Menu Effects and Choice Overload ii

- **Retirement plans:** too many fund options can discourage participation or push people towards simplified decision rules

Takeaway: choices are shaped not only by preferences and prices, but also by how options are organised and presented.

Social Pressure and Persuasion i

People deviate from their own preferences under social pressure and persuasion:

Classic lab evidence:

- **Asch (1951):** 1 in 3 subjects give the *wrong* answer about line length when confederates respond unanimously. Even when the correct answer is clear, social consensus overrides it.
- **Milgram (1963):** 62% of subjects administer apparently dangerous electric shocks when instructed to by an authority figure.

Field evidence:

- **Soccer referees** (Garicano et al., 2005): referees award twice as much extra time when the home team is trailing - crowd pressure, not the clock, is doing the work.

Social Pressure and Persuasion ii

- **Fox News effect** (DellaVigna & Kaplan, 2007): towns where Fox News launched saw a half-percentage-point rise in Republican vote share.
- **Workplace peers** (Mas & Moretti, 2006): when a high-productivity worker joins a shift, their colleagues measurably speed up - even without direct interaction or any financial incentive to do so. Effort is partly social.

Nudge Theory and Libertarian Paternalism

The Key Problem: Should We Do Anything About This?

Two extreme responses to the evidence:

- **Libertarian purist:** "People have revealed preferences; we should respect them even if they look irrational from the outside."
- **Hard paternalist:** "People systematically make mistakes; the state should correct their choices by mandate, ban, or tax."

Thaler & Sunstein's (2003, 2008) proposal: a third way.

The key observation: *any* choice environment has a design. Someone must decide where the fruit goes in the cafeteria. Someone must decide what the default pension option is. Neutrality is impossible - the question is only how to design the environment.

Libertarian Paternalism

Thaler & Sunstein define two concepts:

- **Paternalistic:** a policy is paternalistic if it is chosen with the goal of influencing choices in a way that makes people better off - as measured, as objectively as possible, by their own welfare
- **Libertarian:** the intervention preserves freedom of choice; no options are forbidden; opting out is cheap

Libertarian paternalism: design the choice architecture to promote welfare without coercion.

Key claim (Thaler & Sunstein): Anti-paternalism is incoherent as an absolute position. A cafeteria must arrange food *somehow*. The choice of "no default" is itself a default. The relevant question is: *which* design promotes welfare?

What Is a Nudge?

Definition (Thaler & Sunstein, 2008)
"A nudge . . . is any aspect of the choice architecture that alters people's behaviour in a predictable way without forbidding any options or significantly changing their economic incentives. To count as a mere nudge, the intervention must be easy and cheap to avoid."

Examples:

- Putting fruit at eye level in a cafeteria (nudge) vs. banning desserts (not a nudge)
- Automatic pension enrolment with opt-out (nudge) vs. mandatory contributions (not a nudge)
- Placing a staircase in a prominent position to encourage exercise
- Printing "This is what your neighbours pay" on energy bills

Key question: if people can easily opt out, why does the default matter so much? Answer: present bias, inertia, and status quo bias make defaults "sticky."

Save More Tomorrow (SMarT)

Thaler & Benartzi (2004) - the flagship example of libertarian paternalism in action:

- Employees agree *in advance* to increase their pension contribution every time they get a raise
- No one is forced to participate; opt-out is available at any time
- The design is tailored to present-biased preferences:
  - Commitment is made *in the future* (no immediate cost for the present self)
  - Increases are tied to raises (no nominal income reduction - exploits nominal loss aversion)
- **Results:** average contribution rate rose from 3.5% to 13.6% in four years
- Very few participants opted out

The 2006 Pension Protection Act (US) gave statutory encouragement to plans designed along these lines. **Ireland:** the 2023 Auto-Enrolment Retirement Savings Act follows the same logic.

Evaluating Nudges: The Toolbox

Thaler & Sunstein suggest three criteria for choosing among possible defaults:

1. **Market mimicking:** choose the option the majority would choose if forced to make an explicit decision
2. **Forced choice:** require an active decision when the stakes are high and preferences are clear - eliminate the default
3. **Minimise opt-outs:** choose the option that most people are content to accept - ex post validation of welfare effects

But nudges are not a free lunch:

- Who decides what counts as welfare-enhancing?
- Nudges designed by governments can entrench paternalistic preferences
- Firms can use the same architecture to *exploit* consumers (dark patterns, auto-renewal traps)

Criticisms of Libertarian Paternalism

The libertarian paternalism agenda has been criticised from multiple directions:

- **Glaeser (2006):** governments face the same cognitive biases as individuals, plus additional political economy distortions. Politicians nudge toward re-election, not welfare.
- **Rizzo & Whitman:** there is no stable "true preference" to recover. Biases interact; which one should policy correct?
- **Firms as nudgers:** the same tools used to raise pension savings can be used to keep people subscribed to services they never use. Dark patterns are "reverse nudges."
- **Distributional concerns:** nudges are not equally effective across all groups. More cognitively burdened individuals (poverty, stress) may be more susceptible to manipulation.

How Do Firms Respond to Behavioural Biases? i

DellaVigna & Malmendier (2004) - Behavioural Industrial Organisation:

Profit-maximising firms facing present-biased consumers set prices strategically:

- **Investment goods** (gym): price *below* marginal cost per visit (zero); profit via the lump-sum fee
- **Leisure goods** (credit cards): price *above* marginal cost post-teaser; profit from naive consumers who underestimate future borrowing
- **Gambling:** cheap accommodation and buffets (reduce upfront cost of getting to the casino)

Armstrong & Huck (2010) - Are firms better at this than consumers?

How Do Firms Respond to Behavioural Biases? ii

- Firms *may* be less biased (repeated decisions, market selection, consultants)
- But managers also have biases - overconfidence, short-termism, status-seeking
- Competition does not always eliminate exploitation of biased consumers

Behavioural Economics in an Irish Context i

Where do we see these ideas in Irish policy?

1. **Auto-Enrolment Pensions (2023 Act):** Ireland had one of the lowest voluntary pension coverage rates in the EU. Automatic enrolment with opt-out directly applies Madrian & Shea and the SMarT logic.
2. **Sugar-Sweetened Drinks Tax (2018):** raises the immediate price of a leisure good. A blunter instrument than a nudge, but motivated by the same evidence on present bias and projection bias about future health.
3. **Plain packaging and calorie labelling:** making information harder to ignore is itself a policy lever. Mandatory calorie labelling, plain tobacco packaging, and price comparison websites all work by bringing hidden costs into view - no ban is required.

Behavioural Economics in an Irish Context ii

4. **Rental market regulation:** reference point effects suggest tenants experience rent increases as losses, creating political pressure asymmetric to equivalent gains. This partly explains the politics of rent caps.

Summary: What Does Behavioural Economics Tell Public Choice?

Four big takeaways for this course:

1. **Voters are not homo economicus.** Limited attention, social pressure, and loss framing shape political behaviour in predictable ways - politicians and lobbyists know this.
2. **Institutions shape outcomes even when people can opt out.** Defaults matter. The political fights over pension defaults, benefit take-up, and voter registration are fights over choice architecture.
3. **The market does not always correct bias.** Firms often *amplify* biases when it is profitable to do so - teaser rates, gym contracts, auto-renewal.
4. **Paternalism is not binary.** The Thaler–Sunstein framework replaces a false choice (regulate vs. don't) with a design question: what choice architecture best serves welfare while preserving freedom?

Further Reading

Foundational Papers i

1. Kahneman, D. & Tversky, A. (1979). "Prospect Theory: An Analysis of Decision under Risk." *Econometrica*, 47(2).
2. Tversky, A. & Kahneman, D. (1992). "Advances in Prospect Theory: Cumulative Representation of Uncertainty." *Journal of Risk and Uncertainty*, 5(4).
3. Kahneman, D., Knetsch, J. & Thaler, R. (1990). "Experimental Tests of the Endowment Effect and the Coase Theorem." *JPE*, 98(6).
4. Kahneman, D., Knetsch, J. & Thaler, R. (1986). "Fairness as a Constraint on Profit Seeking." *AER*, 76(4).
5. Thaler, R. (1981). "Some Empirical Evidence on Dynamic Inconsistency." *Economics Letters*.
6. Laibson, D. (1997). "Golden Eggs and Hyperbolic Discounting." *QJE*, 112(2).
7. O'Donoghue, T. & Rabin, M. (1999). "Doing It Now or Later." *AER*, 89(1).
8. Loewenstein, G., O'Donoghue, T. & Rabin, M. (2003). "Projection Bias in Predicting Future Utility." *QJE*, 118(4).

Foundational Papers ii

9. Madrian, B. & Shea, D. (2001). "The Power of Suggestion: Inertia in 401(k) Participation." *QJE*, 116(4).
10. Thaler, R. & Sunstein, C. (2003). "Libertarian Paternalism." *AER P&P*, 93(2).
11. DellaVigna, S. (2007). "Psychology and Economics: Evidence from the Field." *JEL*, 45(2).

Field Evidence i

1. Svenson, O. (1981). "Are We All Less Risky and More Skilful than Our Fellow Drivers?" *Acta Psychologica*, 47(2).
2. Alpert, M. & Raiffa, H. (1982). "A Progress Report on the Training of Probability Assessors." In *Judgment Under Uncertainty: Heuristics and Biases*. Cambridge UP.
3. Forsythe, R. et al. (1994). "Fairness in Simple Bargaining Experiments." *Games and Economic Behavior*, 6(3).
4. Ariely, D. & Wertenbroch, K. (2002). "Procrastination, Deadlines, and Performance." *Psychological Science*, 13(3).
5. Camerer, C. et al. (1997). "Labor Supply of New York City Cab Drivers." *QJE*, 112(2).
6. Terrell, D. (1994). "A Test of the Gambler's Fallacy: Evidence from Pari-Mutuel Games." *Journal of Risk and Uncertainty*, 8(3).
7. Odean, T. (1999). "Do Investors Trade Too Much?" *AER*, 89(5).
8. Malmendier, U. & Tate, G. (2005). "CEO Overconfidence and Corporate Investment." *Journal of Finance*, 60(6).

Field Evidence ii

9. List, J. (2003). "Does Market Experience Eliminate Market Anomalies?" *QJE*, 118(1).
10. Read, D. & van Leeuwen, B. (1998). "Predicting Hunger: The Effects of Appetite and Delay on Choice." *Organizational Behavior and Human Decision Processes*, 76(2).
11. Conlin, M., O'Donoghue, T. & Vogelsang, T. (2007). "Projection Bias in Catalog Orders." *AER*, 97(4).
12. Fehr, E. & Goette, L. (2007). "Do Workers Work More if Wages Are High?" *AER*, 97(1).
13. Card, D. & Hyslop, D. (1997). "Does Inflation Grease the Wheels of the Labor Market?" In *Reducing Inflation*. NBER/University of Chicago Press.
14. Gneezy, U. & List, J. (2006). "Putting Behavioral Economics to Work." *Econometrica*, 74(5).
15. Hossain, T. & Morgan, J. (2006). "...Plus Shipping and Handling." *Advances in Economic Analysis & Policy*, 6(2).

Field Evidence iii

16. Chetty, R., Looney, A. & Kroft, K. (2009). "Salience and Taxation: Theory and Evidence." *AER*, 99(4).
17. Iyengar, S., Huberman, G. & Jiang, W. (2004). "How Much Choice Is Too Much?" In *Pension Design and Structure*. Oxford UP.
18. Ho, D. & Imai, K. (2008). "Estimating Causal Effects of Ballot Order from a Randomized Natural Experiment." *Public Opinion Quarterly*, 72(2).
19. Garicano, L., Palacios-Huerta, I. & Prendergast, C. (2005). "Favoritism Under Social Pressure." *Review of Economics and Statistics*, 87(2).
20. DellaVigna, S. & Kaplan, E. (2007). "The Fox News Effect." *QJE*, 122(3).
21. Mas, A. & Moretti, E. (2009). "Peers at Work." *AER*, 99(1).
22. Benartzi, S. (2001). "Excessive Extrapolation and the Allocation of 401(k) Accounts to Company Stock." *Journal of Finance*, 56(5).

On Nudges, Firms, and Policy Design i

1. Thaler, R. & Benartzi, S. (2004). "Save More Tomorrow." *JPE*, 112(S1).
2. Thaler, R. & Sunstein, C. (2008). *Nudge*. Yale University Press.
3. Glaeser, E. (2006). "Paternalism and Psychology." *University of Chicago Law Review*, 73(1).
4. Camerer, C. et al. (2003). "Regulation for Conservatives: Asymmetric Paternalism." *UPenn Law Review*, 151(3).
5. DellaVigna, S. & Malmendier, U. (2004). "Contract Design and Self-Control." *QJE*, 119(2).
6. DellaVigna, S. & Malmendier, U. (2006). "Paying Not to Go to the Gym." *AER*, 96(3).
7. Gabaix, X. & Laibson, D. (2006). "Shrouded Attributes, Consumer Myopia, and Information Suppression." *QJE*, 121(2).
8. Armstrong, M. & Huck, S. (2010). "Behavioral Economics as Applied to Firms: A Primer." *Competition Policy International*, 6(1).
9. Benartzi, S. & Thaler, R. (1995). "Myopic Loss Aversion and the Equity Premium Puzzle." *QJE*, 110(1).

On Nudges, Firms, and Policy Design ii

10. Benartzi, S. & Thaler, R. (2001). "Naive Diversification Strategies in Defined Contribution Savings Plans." *AER*, 91(1).
11. Camerer, C. & Lovallo, D. (1999). "Overconfidence and Excess Entry." *AER*, 89(1).
12. Haigh, M. & List, J. (2005). "Do Professional Traders Exhibit Myopic Loss Aversion?" *Journal of Finance*, 60(1).
13. Laibson, D. et al. (2007). "Estimating Discount Functions with Consumption Choices over the Lifecycle." NBER Working Paper 13314.
14. Mehra, R. & Prescott, E. (1985). "The Equity Premium: A Puzzle." *Journal of Monetary Economics*, 15(2).

Irish and European Policy Context

1. OECD (2017). *Behavioural Insights and Public Policy: Lessons from Around the World*. OECD Publishing.
2. Houses of the Oireachtas (2023). *Automatic Enrolment Retirement Savings System Act*.
3. Revenue Commissioners / ESRI (2018). *The Sugar-Sweetened Drinks Tax: First-Year Evidence*. ESRI Working Paper.
4. DETE (2022). *Consumer Protection in Digital Markets: Discussion Paper*. Dept of Enterprise.
5. Lunn, P. (2014). *Regulatory Policy and Behavioural Economics*. OECD. (Pete Lunn is at the ESRI - directly relevant to Irish context)

Cultural Resources

Films and Documentaries

Behavioural Economics in the Wild:

- **The Big Short** (2015) - overconfidence, herding, and limited attention in financial markets; the "Jenga" scene is practically a seminar in behavioural finance
- **Supersize Me** (2004) - present bias, projection bias, and the food industry's exploitation of both
- **Inside Job** (2010) - overconfidence, regulatory capture, and what happens when rational institutions interact with biased markets
- **Parasite** (2019) - framing, reference dependence, and the psychology of relative deprivation

Books for General Audience

- Richard Thaler & Cass Sunstein - *Nudge* (2008, updated 2021) - the manifesto
- Daniel Kahneman - *Thinking, Fast and Slow* (2011) - the most accessible synthesis of the cognitive bias literature; written by one of the architects
- Dan Ariely - *Predictably Irrational* (2008) - engaging field experiments on irrationality; some replication caveats apply
- Richard Thaler - *Misbehaving* (2015) - a memoir of the behavioural economics revolution; unusually candid
- Michael Lewis - *The Undoing Project* (2016) - the story of Kahneman and Tversky's collaboration; reads like a novel

Podcasts and Online Resources

Behavioural Insights in Practice:

- **Freakonomics Radio** - "How to Be More Productive" and multiple episodes on nudging
- **EconTalk** - episodes with Thaler, Sunstein, and List
- **Planet Money** - "The Troubled Bridge Over Water" and episodes on defaults and savings
- **The Nudge Podcast** - dedicated to applied behavioural insights

Data and Evidence:

- **ESRI Behavioural Research Unit** (`esri.ie`) - Irish applied work by Pete Lunn's team
- **ideas42.org** - applied behavioural design in development contexts
- **BIT (Behavioural Insights Team)** - the original UK "nudge unit": case studies and reports

Backup Slides

The (β, δ) Model of Present Bias (Formal) i

For those interested in the formal structure:

Standard discounting: $U_t = u_t + \delta u_{t+1} + \delta^2 u_{t+2} + \cdots$

Present-bias model (Laibson 1997; O'Donoghue & Rabin 1999):

$$U_t = u_t + \beta\delta u_{t+1} + \beta\delta^2 u_{t+2} + \cdots \quad \beta \leq 1$$

- δ: how much you discount one period into the future (same as always)
- β: an *extra* penalty applied to everything that is not the present moment
- $\beta = 1$: standard model; $\beta < 1$: the present is over-weighted

The (β, δ) Model of Present Bias (Formal) ii

Investment good ($b_1 < 0$, $b_2 > 0$): the present-biased agent feels the cost more acutely, so does less exercise, less saving, less studying than she planned.

Leisure good ($b_1 > 0$, $b_2 < 0$): she over-consumes relative to her own long-run preferences.

Estimated values: Laibson et al. (2006): $\beta \approx 0.70$, $\delta \approx 0.96$ from life-cycle savings data.

Behavioural Economics: A Brief Intellectual History

- **1955** - Herbert Simon proposes "bounded rationality"; agents use heuristics
- **1974** - Tversky & Kahneman: "Judgment under Uncertainty: Heuristics and Biases" (*Science*)
- **1979** - Kahneman & Tversky: Prospect Theory (*Econometrica*) - the most cited paper in economics since 1970
- **1981** - Thaler documents dynamic inconsistency in intertemporal choice
- **1997** - Laibson formalises the (β, δ) model
- **2001** - Madrian & Shea: default effects in 401(k)s
- **2002** - Kahneman awarded the Nobel Prize in Economics
- **2003** - Thaler & Sunstein: "Libertarian Paternalism"
- **2008** - *Nudge* published; UK establishes Behavioural Insights Team (2010)
- **2017** - Thaler awarded the Nobel Prize in Economics

Are Firms Smarter Than Consumers?

Armstrong & Huck (2010) identify reasons firms might be less biased:

- Economies of scale in decision-making: firms face the same decision repeatedly
- Market selection: consistently bad decision-making firms exit
- Access to consultants and data; internal checks and governance

But firms also exhibit systematic biases:

- Overconfident CEOs overpay for mergers (Malmendier & Tate)
- Agency problems: managers maximise their own status, not shareholder value
- Complex environments generate rules of thumb that may calcify
- "Crazy" strategic behaviour can be rational individually but wasteful socially

Exit is not instantaneous: firms often decline for long periods before failure, so selection is slow and imperfect.

The Equity Premium Puzzle i

A famous puzzle in finance that behavioural economics helps resolve:

- US equity returns outperformed bonds by ≈ 3.9 percentage points per year (1871–1993)
- Under the standard model, this requires implausibly high risk aversion (Mehra & Prescott, 1985)

Benartzi & Thaler (1995) - Myopic Loss Aversion:

- If investors evaluate portfolio performance *annually*, the probability of stocks underperforming bonds in any given year is substantial
- Loss-averse investors (with $\lambda \approx 2.25$) require a substantial equity premium to tolerate this risk
- Calibration matches the observed premium without requiring extreme risk aversion

The Equity Premium Puzzle ii

Key insight: the combination of loss aversion and short evaluation horizons ("myopic loss aversion") can explain what standard models cannot.

Experience, Markets, and the Limits of Debiasing

A common objection: "Won't markets and experience correct these biases?"

DellaVigna's four responses:

1. Many important decisions are *infrequent* with noisy feedback (house purchase, pension choice) - learning cannot occur
2. Experience can *reinforce* some biases (professional investors show *more* myopic loss aversion than students: Haigh & List, 2004)
3. Firms typically have *no incentive to debias* consumers when bias is profitable (Gabaix & Laibson, 2006)
4. Some preferences (social preferences, loss aversion) are not "errors" to be corrected by experience - they are deep features of preferences

Bottom line: market competition is necessary but not sufficient for debiasing. The cases where it works well (experienced traders, commodity markets) are precisely where S&D performs well already.